

ORIGINAL ARTICLE

Early outcomes of laparoscopic totally extraperitoneal repair with protack mesh fixation for bilateral inguinal hernia

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ABSTRACT

Background: Bilateral inguinal hernia accounts for 15-20% of all inguinal hernias. The role of mesh fixation during laparoscopic totally extraperitoneal (TEP) repair for bilateral inguinal hernia remains controversial. This study evaluated the clinical characteristics, European Hernia Society (EHS) classification, and early outcomes of laparoscopic TEP repair with ProTack mesh fixation for bilateral inguinal hernia.

Methods: A prospective descriptive study included 45 patients with bilateral inguinal hernia who underwent laparoscopic TEP repair with ProTack mesh fixation at Can Tho General Hospital between July 2025 and May 2026.

Results: The mean age was 64.8 ± 10.9 years, and all patients were male. Direct hernias accounted for 65.6% of all hernia sacs, while M2 was the most common EHS subtype (46.7%). The mean operative time was 67.1 ± 16.8 minutes. Peritoneal tears and spermatic vessel injuries occurred in 35.6% and 13.3% of patients, respectively. Mean VAS scores decreased from 2.93 ± 0.86 at 24 hours to 1.33 ± 0.50 at 72 hours. Mean postoperative hospital stay was 1.58 ± 0.78 days. Early complications included seroma (11.1%) and urinary retention (4.4%). At follow-up, only one patient (3.2%) reported mild groin paresthesia. No conversion, chronic pain, mesh infection, or recurrence was observed. The overall success rate was 100%.

Conclusions: Laparoscopic TEP repair with ProTack mesh fixation demonstrated favorable early clinical outcomes and appears to be a feasible and safe approach for bilateral inguinal hernia. (www.actabiomedica.it)

Key words: bilateral inguinal hernia, totally extraperitoneal repair, TEP, mesh fixation, ProTack



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Introduction

Inguinal hernia is one of the most common conditions encountered in general surgery, accounting for approximately 75% of all abdominal wall hernias. Bilateral inguinal hernia represents about 15–20% of inguinal hernia cases and is more frequently observed in elderly patients and individuals exposed to chronic increases in intra-abdominal pressure (1-3). Surgical repair remains the only definitive treatment, with more than 20 million inguinal hernia operations performed worldwide each year (1).

Over the last three decades, laparoscopic inguinal hernia repair has gained widespread acceptance because of its advantages of reduced postoperative pain, faster recovery, earlier return to normal activities, and improved cosmetic outcomes compared with conventional open repair (4-6). Among laparoscopic techniques, totally extraperitoneal (TEP) repair has become a preferred approach because it avoids entry into the peritoneal cavity while allowing mesh placement within the preperitoneal space to reinforce the entire myopectineal orifice (4,5). In addition, TEP is particularly suitable for bilateral inguinal hernia because both groins can be repaired through the same access without additional incisions. The European Hernia Society (EHS) classification has been widely adopted to standardize hernia assessment and facilitate comparisons among studies (7).

Despite the increasing popularity of TEP repair, the role of mesh fixation remains controversial. Although mesh fixation may reduce the risk of mesh displacement and recurrence, particularly in large direct hernias, it may also increase operative costs and postoperative pain related to tissue or nerve injury (8-10). This issue is of particular interest in bilateral inguinal hernia repair, where extensive bilateral preperitoneal dissection creates a larger potential space that may increase the risk of mesh migration or folding. Previous studies comparing fixation and non-fixation techniques have reported conflicting results, and evidence specifically focusing on bilateral inguinal hernia remains limited (10,11). To the best of our knowledge, no previous Vietnamese study has specifically evaluated laparoscopic TEP repair with ProTack mesh fixation for bilateral inguinal hernia. Therefore, the present

study may represent the first report from Vietnam focusing on the clinical characteristics and early outcomes of this technique.

Therefore, this study aimed to evaluate the clinical characteristics, EHS classification, and early outcomes of laparoscopic TEP repair with ProTack mesh fixation for bilateral inguinal hernia at Can Tho General Hospital.

Materials and Methods

Study subjects

A prospective descriptive study was conducted at the Department of General Surgery, Can Tho General Hospital, Vietnam, from July 2025 to May 2026. The study included consecutive adult patients diagnosed with bilateral inguinal hernia who underwent laparoscopic totally extraperitoneal (TEP) repair with mesh fixation using ProTack.

Inclusion and exclusion criteria

Patients were eligible if they had primary bilateral inguinal hernia, were classified as American Society of Anesthesiologists (ASA) physical status I–III, and agreed to participate in the study.

Patients were excluded if they had a history of lower abdominal surgery, previous TEP repair, pelvic irradiation, active pelvic or systemic infection, or recurrent inguinal hernia previously treated by the TEP approach.

Surgical technique

All procedures were performed under general anesthesia using the laparoscopic TEP approach. After creation of the preperitoneal space, dissection was carried out to expose the myopectineal orifice bilaterally. Hernia sacs were reduced and the preperitoneal space was adequately prepared. A polypropylene mesh measuring 10 × 15 cm was placed on each side to cover the entire myopectineal orifice. Mesh fixation was performed using titanium ProTack devices. Typically, three to five tacks were placed to secure each mesh, including fixation to Cooper's ligament over the pubic

tubercle and the anterior abdominal wall superiorly, while avoiding the triangle of pain and triangle of doom. The procedure was completed after confirming appropriate mesh positioning and hemostasis.

Data collection and Outcome measures

Demographic and clinical variables included age, sex, body mass index (BMI), occupation, risk factors, duration of symptoms, and ASA classification. Hernias were classified according to the European Hernia Society (EHS) classification system.

Operative outcomes included operative time, intraoperative findings, and surgical complications. Postoperative outcomes included pain assessed using the Visual Analog Scale (VAS), recovery time, length of hospital stay, early postoperative complications, patient satisfaction and treatment success. Treatment success was defined as successful completion of laparoscopic TEP repair with ProTack mesh fixation without conversion to open surgery, absence of major postoperative complications requiring reoperation, and no hernia recurrence during the available follow-up period.

Postoperative analgesia consisted of intravenous paracetamol during the first 24 hours, followed by oral nonsteroidal anti-inflammatory drugs as needed according to the patient's pain intensity.

Patient satisfaction was assessed at the latest available follow-up visit using a simple two-category assessment (completely satisfied or not completely satisfied) based on the patient's overall evaluation of the surgical outcome.

Follow-up

Patients were evaluated during hospitalization and subsequently followed at 1 month, 3 months, and 6 months after surgery. As patient enrollment continued until the end of the study period, some patients had not yet reached the scheduled 3- and 6-month follow-up time points when the data were analyzed. Follow-up assessments focused on postoperative complications, chronic groin pain, sensory disturbances, mesh-related complications, and hernia recurrence.

Statistical analyses

Data were analyzed using SPSS software. Continuous variables are presented as mean $\pm$ standard deviation, whereas categorical variables are presented as frequencies and percentages.

Ethical considerations

The study was approved by the institutional ethics committee of Can Tho University of Medicine and Pharmacy and conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants.

Results

A total of 45 patients with bilateral inguinal hernia were included in this study. All patients underwent laparoscopic TEP repair with bilateral polypropylene mesh placement and ProTack fixation. No patient required conversion to open surgery, and no major intraoperative visceral or vascular injuries were recorded.

The majority of patients were elderly men with associated risk factors such as smoking and benign prostatic hyperplasia. Direct hernias were more common than indirect hernias, and M2 defects represented the largest subgroup according to the EHS classification. The mean operative time was 67.1 ± 16.8 minutes. Postoperative recovery was rapid, with a mean hospital stay of 1.58 ± 0.78 days and progressively decreasing pain scores during the first 72 hours. Early postoperative complications were infrequent and mostly self-limiting. Follow-up data were available for 31 patients at 3 months and 17 patients at 6 months. Only one patient (3.2%) reported mild groin paresthesia at 3 months, which resolved during subsequent follow-up. No chronic pain, mesh infection, or hernia recurrence was observed during the follow-up period.

Detailed patient characteristics, hernia classification, operative findings, and postoperative outcomes are summarized in Tables 1-4.

Most patients were elderly males. ASA I-II accounted for 84.4% of cases. Smoking and benign prostatic hyperplasia were the most common risk factors.

Table 1. Patient characteristics (n = 45).

Characteristics	Value
Age (years), mean $\pm$ SD	64.8 $\pm$ 10.9
Age >60 years, n (%)	34 (75.6)
Male sex, n (%)	45 (100)
BMI (kg/m ²), mean $\pm$ SD	22.1 $\pm$ 2.3*
Underweight, n (%)	4 (8.9)
Normal BMI, n (%)	27 (60.0)
Overweight, n (%)	12 (26.7)
Obesity, n (%)	2 (4.4)
ASA I, n (%)	18 (40.0)
ASA II, n (%)	20 (44.4)
ASA III, n (%)	7 (15.6)
Current/former smoker, n (%)	21 (46.7)
Benign prostatic hyperplasia, n (%)	16 (35.6)
Chronic constipation, n (%)	9 (20.0)

*Calculated from BMI categories.

Table 2. Hernia characteristics and EHS classification.

Hernia characteristics	n (%)
Total hernias analyzed	90
Direct hernias, n (%)	59 (65.6)
Indirect hernias, n (%)	31 (34.4)
EHS L1, n (%)	14 (15.6)
EHS L2, n (%)	13 (14.4)
EHS L3, n (%)	4 (4.4)
EHS M1, n (%)	7 (7.8)
EHS M2, n (%)	42 (46.7)
EHS M3, n (%)	10 (11.1)

Table 3. Intraoperative outcomes.

Intraoperative Outcomes	Value
Operative time (min), mean $\pm$ SD	67.1 $\pm$ 16.8
Range (min)	40–120
Peritoneal tear, n (%)	16 (35.6)
Spermatic vessel injury, n (%)	6 (13.3)
Visceral injury, n (%)	0
Conversion to open surgery, n (%)	0
Omental content in hernia sac, n (%)	10 (22.2)
Bowel-containing hernia sac, n (%)	1 (2.2)

Table 4. Early postoperative outcomes and follow-up.

Outcomes	Value
VAS score at 24 h, mean $\pm$ SD	2.93 $\pm$ 0.86
VAS score at 48 h, mean $\pm$ SD	2.12 $\pm$ 0.49
VAS score at 72 h, mean $\pm$ SD	1.33 $\pm$ 0.50
Recovery time <12 h, n (%)	33 (73.3)
Postoperative hospital stay (days), mean $\pm$ SD	1.58 $\pm$ 0.78
Seroma, n (%)	5 (11.1)
Urinary retention, n (%)	2 (4.4)
Groin swelling, n (%)	1 (2.2)
Subcutaneous emphysema, n (%)	1 (2.2)
Groin paresthesia at 3 months, n (%)	1 (3.2)
Chronic pain, n (%)	0
Mesh infection, n (%)	0
Recurrence, n (%)	0
Follow-up at 3 months, n (%)	31 (68.9)
Follow-up at 6 months, n (%)	17 (37.8)
Overall treatment success, n (%)	45 (100)
Complete patient satisfaction, n (%)	44 (97.8)

Direct hernias predominated, accounting for nearly two-thirds of all hernia defects. M2 was the most common EHS subtype.

All procedures were completed laparoscopically. Peritoneal tear was the most common intraoperative event.

Based on the predefined criteria, the overall treatment success rate was 100%. Postoperative pain was mild and decreased progressively during follow-up. Early complications were uncommon and mostly self-limiting. No chronic pain, mesh infection, or hernia recurrence was observed during follow-up.

Discussion

Patient characteristics

The present study showed that bilateral inguinal hernia predominantly affected elderly men, with a mean age of 64.8 $\pm$ 10.9 years and 75.6% of patients

older than 60 years. These findings are consistent with the well-established association between aging and progressive weakening of the abdominal wall, particularly the transversalis fascia and posterior wall of the inguinal canal (1-3). Similar age distributions have been reported in previous studies of laparoscopic TEP repair. Nguyen Van Phuoc (12), Nguyen Hoang Ngan Tuan (13), and La Van Phu (14) also found that most patients undergoing TEP repair were older than 50 years.

All patients in the present study were male. Our results are consistent with the known epidemiology of inguinal hernia and with previous Vietnamese and international reports demonstrating a marked male predominance because of anatomical differences in the inguinal canal and embryologic development of the processus vaginalis (1-3, 12-14). Smoking, benign prostatic hyperplasia, and chronic constipation were common risk factors in our cohort. These conditions may contribute to chronic increases in intra-abdominal pressure and have been recognized as important factors in the development of inguinal hernia (1-3).

Hernia characteristics and EHS classification

A notable finding of the present study was the predominance of direct hernias, which accounted for 65.6% of all hernia defects. This observation may be explained by the relatively advanced age of the study population because direct hernias are strongly associated with acquired weakening of the posterior wall of the inguinal canal (1-3). Hidalgo et al. (11) similarly reported that direct defects become increasingly common in patients with bilateral inguinal hernia, particularly among elderly individuals.

The European Hernia Society (EHS) classification has become the most widely accepted system for describing groin hernias and facilitates comparison among studies (7). In our series, M2 defects represented 46.7% of all hernia defects and constituted the most common subtype. This finding is clinically relevant because larger medial defects are often considered situations in which mesh fixation may provide additional stability and potentially reduce the risk of mesh displacement (5,8,9).

Intraoperative outcomes

The mean operative time in our study was 67.1 ± 16.8 minutes, which is comparable to previously published TEP series (5,11,15). Ghazzal et al. (15) reported operative times ranging from 60 to 80 minutes during the initial experience with TEP repair, whereas Hidalgo et al. (11) reported similar operative durations for bilateral laparoscopic hernia repair. Although bilateral TEP repair generally requires longer operative times than unilateral repair because dissection and mesh placement must be performed on both sides, the ability to repair both groins through the same access remains one of the major advantages of the laparoscopic approach (4,5).

Peritoneal tear was the most common intraoperative event, occurring in 35.6% of patients. Although the incidence appears relatively high, peritoneal injury is a well-recognized event during TEP repair and is particularly common during the learning phase or when extensive bilateral dissection is required (15,16). Koyama et al. (16) emphasized that peritoneal tears remain one of the most frequent technical challenges during TEP repair and may compromise the preperitoneal working space if not managed appropriately. In our study, the relatively high incidence of peritoneal tears may also be related to the need for extensive bilateral preperitoneal dissection. Furthermore, most patients were elderly, and age-related tissue fragility may have increased the susceptibility to inadvertent peritoneal injury during dissection. Importantly, all peritoneal tears in our study were successfully managed laparoscopically without conversion to open surgery or postoperative sequelae.

Spermatic vessel injury occurred in 13.3% of patients. All injuries consisted of limited bleeding from the spermatic vessels during cord dissection and were controlled intraoperatively without adverse consequences. No patient required orchiectomy or developed testicular complications during follow-up, indicating that meticulous dissection can minimize the clinical impact of these injuries.

Early postoperative outcomes

Postoperative pain was generally mild in our study. Mean VAS scores progressively decreased from

2.93 ± 0.86 at 24 hours to 1.33 ± 0.50 at 72 hours. These findings are consistent with previous studies demonstrating that laparoscopic TEP repair is associated with less postoperative pain and faster recovery than conventional open repair (4-6). The minimally invasive nature of TEP repair avoids extensive tissue disruption and may contribute to improved postoperative comfort.

The mean postoperative hospital stay was only 1.58 ± 0.78 days. Similar short hospital stays have been reported in both international and Vietnamese studies evaluating TEP repair (11-14). Early postoperative complications were uncommon. Seroma occurred in 11.1% of patients and represented the most frequent complication. This complication is generally self-limiting and is particularly common in patients with large direct hernias because of the residual dead space after reduction of the hernia sac (5,15).

Urinary retention occurred in 4.4% of patients and resolved with conservative management. At follow-up, only one patient (3.2%) reported mild groin paresthesia. No chronic pain, mesh infection, or recurrence was observed. Chronic postoperative pain remains one of the most important concerns following inguinal hernia repair, particularly when mechanical fixation devices are used (10). Reinbold (10) identified fixation-related nerve irritation as an important risk factor for chronic pain. The absence of chronic pain in the present study may be attributed to careful mesh fixation, meticulous dissection, and avoidance of fixation within the triangle of pain.

Role of mesh fixation in bilateral TEP repair

Whether mesh fixation is necessary during TEP repair remains one of the most debated issues in modern hernia surgery (8-10). Several randomized studies and meta-analyses have suggested that routine fixation may not be required in many patients undergoing unilateral TEP repair because recurrence rates are comparable and postoperative pain may be reduced when fixation is omitted (8,9).

However, bilateral inguinal hernia presents a different surgical scenario. During bilateral TEP repair, extensive preperitoneal dissection is required on both sides, creating a substantially larger operative

space than in unilateral procedures. Under these circumstances, some surgeons advocate mesh fixation to reduce the risk of mesh migration, folding, or displacement before tissue incorporation occurs (5,8,9). This concern may be particularly relevant in patients with large medial defects such as M2 or M3 hernias, which accounted for a substantial proportion of our study population.

Ayyaz et al. (8) reported favorable outcomes with mesh fixation and suggested that fixation may reduce the risk of recurrence in selected patients. Similarly, Techapongsatorn et al. (9), in a systematic review and meta-analysis, concluded that current evidence remains insufficient to completely abandon mesh fixation in all TEP repairs. Evidence specifically focusing on bilateral inguinal hernia remains limited. Hidalgo et al. (11) highlighted the need for additional studies evaluating optimal mesh management strategies in bilateral laparoscopic hernia repair.

In the present study, ProTack fixation was routinely used, and no recurrence or mesh-related complication was observed during follow-up. Although our study was not designed to compare mesh fixation with non-fixation or other repair techniques, the favorable outcomes observed in this cohort suggest that laparoscopic TEP repair with ProTack mesh fixation is a feasible and safe approach for bilateral inguinal hernia. However, this prospective descriptive study does not permit conclusions regarding the comparative effectiveness or clinical superiority of mesh fixation. Further prospective comparative studies with larger sample sizes and longer follow-up are required to determine whether routine mesh fixation offers additional benefits over non-fixation or other repair strategies.

Conclusion

In this prospective descriptive study, laparoscopic TEP repair with ProTack mesh fixation demonstrated favorable early clinical outcomes, including low postoperative pain, short hospital stay, and high patient satisfaction. These findings suggest that laparoscopic TEP repair with ProTack mesh fixation is a feasible and safe approach in our cohort. However, comparative studies are required to determine whether mesh

fixation provides additional clinical benefit over non-fixation or other repair strategies.

Limitations

Several limitations should be acknowledged. This was a single-center study with a relatively small sample size and a limited follow-up duration, limiting the assessment of long-term outcomes such as recurrence and chronic postoperative pain. Because patient enrollment continued throughout the study period, not all patients had reached the scheduled follow-up time points at the time of data analysis. Furthermore, the absence of a comparison group meant that this descriptive study could only report the outcomes of ProTack mesh fixation and could not determine its comparative effectiveness over non-fixation or other repair strategies. Nevertheless, this study provides valuable preliminary data on the early outcomes of laparoscopic TEP repair with ProTack mesh fixation for bilateral inguinal hernia. Further comparative studies with larger sample sizes and longer follow-up are warranted.

Ethic Approval: The study protocol was reviewed and approved by the Ethics Committee for Biomedical Research of Can Tho University of Medicine and Pharmacy (Approval No. 25.484. HV/PCT-HĐĐĐ, approved on June 30, 2025). All procedures were conducted in accordance with the ethical principles of the Declaration of Helsinki. Written informed consent was obtained from all participants before enrollment.

Conflict of Interest: The authors declare no conflicts of interest.

Authors' Contribution: Duong Tuan Tu contributed to the study conceptualization, data collection, data curation, formal analysis, investigation, methodology, and preparation of the original draft. Hong Quan Dang contributed to methodology, supervision, data interpretation, and critical revision of the manuscript. La Van Phu contributed to study design, project administration, supervision, validation of the results, manuscript revision, and final approval of the manuscript. All authors reviewed and approved the final version of the manuscript and agree to be accountable for all aspects of the work.

Declaration on the use of AI: None.

Data Sharing Statement: Raw data were generated at Can Tho General Hospital. Derived data supporting the findings of this study are available from the corresponding author on request.

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